

Project Design Phase
Proposed Solution Template

Date	29 September 2026
Team ID	SWTID-2026-8776
Project Name	AI WeatherWise API
Maximum Marks	2 Marks

Chevalier T. Thomas Elizabeth College for Women

Team:

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Proposed Solution Template:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Users often struggle to interpret raw weather data and lack personalized recommendations for travel, clothing, and outdoor activities. Managing multiple weather locations securely is also challenging.
2.	Idea / Solution description	AI WeatherWise API is a RESTful backend platform that integrates real-time weather data with Google Gemini AI to generate intelligent weather summaries and personalized recommendations. It also provides secure user authentication and favorite-location management.
3.	Novelty / Uniqueness	Combines real-time weather information with AI-generated natural language summaries and personalized recommendations in a single platform. The system transforms complex weather metrics into practical and easy-to-understand insights.
4.	Social Impact / Customer Satisfaction	Helps users make informed decisions regarding travel, outdoor activities, and daily planning. Improves user experience by providing personalized weather guidance and reducing confusion caused by raw weather data.
5.	Business Model (Revenue Model)	Freemium model offering basic weather services for free, while premium users can access advanced AI insights, detailed forecasts, and additional personalized recommendation features. API subscription plans can also be offered to third-party applications.
6.	Scalability of the Solution	The system is built using Node.js, Express.js, MongoDB, and REST APIs, allowing it to support increasing numbers of users, locations, and AI requests. Additional weather providers, forecasting features, and frontend applications can be integrated in the future.

